

Exact Series and Error-Controlled Evaluation of the Nonconvex ℓ_p Proximity Operator

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Abstract

We study the scalar proximity operator of the nonconvex power penalty with exponent between zero and one. Lagrange–Bürmann inversion gives an exact series for the larger stationary root, and the series converges uniformly throughout the active domain of the proximity operator. For rational exponents, an explicit residue-class decomposition reorganizes the coefficients into finite sums of generalized hypergeometric functions, recovering the familiar special cases and producing an additional formula for $p = \frac{1}{3}$. A Mellin–Barnes formula yields an equivalent residue-corrected representation and analytic truncation bounds. We distinguish the proximal activation threshold from the lower saddle-node threshold and prove that the active proximal branch is uniformly well-conditioned; the practical difficulty near activation is slow series convergence when the exponent approaches one, rather than singular root sensitivity. We also formulate an inexact proximal-gradient step using a computable optimality residual and a descent test. With summable descent errors and vanishing residuals, the objective values converge, the steps vanish, and every cluster point is critical. Tests over a broad parameter grid and in sparse-recovery proximal-gradient runs illustrate both evaluator accuracy and the implementability of the acceptance certificates.

Keywords: Proximity operator; nonconvex optimization; Lagrange–Bürmann inversion; Mellin–Barnes integral; inexact proximal gradient.

2020 Mathematics Subject Classification: 90C26, 65K10, 49J52, 33C20.

1 Introduction

Sparsity priors are central to high-dimensional learning and inverse problems. The ℓ_0 penalty models sparsity directly but leads to combinatorial optimization, whereas the convex ℓ_1 relaxation is computationally attractive but can over-shrink large coefficients [7, 8]. The nonconvex ℓ_p penalty with $0 < p < 1$ lies between these choices: it promotes sparsity more strongly than ℓ_1 while reducing bias, but it also complicates analysis and computation [9, 11, 16, 21, 30]. Moreover, empirical studies indicate that natural-image wavelet coefficients are often well modeled by p -values in the range $[0.5, 0.8]$ [29].

A core primitive in first-order methods for nonsmooth regularization is the proximity operator (see, e.g., [2, 4, 13, 23, 27]). Recent work has also emphasized explicit and structured computation

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of proximity operators for broad classes of penalties [6, 19]. For a function f , the operator is defined by

$$\text{prox}_{\lambda f}(y) \in \arg \min_x \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda f(x) \right\},$$

and it reduces componentwise to a scalar problem when $f(x) = \|x\|_p^p$. Hard- and soft-thresholding give closed forms for ℓ_0 and ℓ_1 , and formulas are also known for $p = \frac{1}{2}$ and $p = \frac{2}{3}$ [10, 12]. For a general exponent, Newton and bisection are available [12, 24]; our purpose is to provide a uniform analytic representation and an alternative evaluation mechanism with transparent truncation error.

The scalar ℓ_p proximity operator is governed by the input activation threshold $\tau_{p,\lambda}$ and the corresponding nonzero output level $\rho_{p,\lambda}$, both defined explicitly in Section 2. For $|y| > \tau_{p,\lambda}$, its value is $\text{sgn}(y)x_+(|y|)$, where $x_+(|y|)$ is the larger positive root of

$$x - |y| + \lambda p x^{p-1} = 0, \quad x > 0.$$

At $|y| = \tau_{p,\lambda}$, both zero and $\text{sgn}(y)\rho_{p,\lambda}$ are minimizers. Thus, evaluating the nonzero output amounts to following one specified inverse branch rather than finding an arbitrary stationary point.

When $p = r/s \in (0, 1)$ is rational and written in lowest terms, the positive stationarity equation can be transformed into a trinomial algebraic equation. Classical and recent power-series and hypergeometric representations of related trinomial roots [17, 26], together with Mellin–Barnes representations of trinomial algebraic functions [20], therefore provide relevant background. Our contribution is not the mere existence of such representations in the rational case. Rather, we identify the real branch selected by the global proximal minimizer, derive a unified inverse-series and Mellin–Barnes framework valid for every real $p \in (0, 1)$, including irrational p , prove uniform convergence on the full active proximal domain, and connect the evaluation error to computable conditions for an inexact proximal-gradient method.

We recast the normalized stationarity equation as an analytic inverse problem near the regular point $v = 1$. The main contributions are as follows.

- Lagrange–Bürmann inversion yields an exact power series for $x_+(y)$ for every $0 < p < 1$. The series converges uniformly on the entire active domain of the proximity operator.
- For every rational exponent, we give an explicit residue-class formula that reduces the inverse series to a finite sum of generalized hypergeometric functions. This recovers the classical cases and gives a compact representation for $p = \frac{1}{3}$.
- An equivalent Mellin–Barnes representation leads to a residue-corrected finite-contour evaluator. We separate analytic tail error from numerical quadrature error, so the strength of the output guarantee is stated explicitly.
- We distinguish the proximal activation threshold from the lower saddle-node threshold. The active proximal branch satisfies a uniform condition-number bound, while the worst active-domain series convergence factor approaches one only as $p \uparrow 1$.
- For proximal-gradient algorithms, we replace a distance-only inexactness condition by a computable optimality residual and a descent test. With summable descent errors and vanishing residuals, objective values converge, steps vanish, and every cluster point is critical.

Section 2 reviews the threshold structure. Section 3 derives the inverse series, Section 4 gives rational-exponent representations, and Section 5 develops the Mellin–Barnes formula, the evaluator, and its use in proximal gradient. Section 6 concludes. A separate supplement gives a Galois-theoretic classification of radical formulas.

2 Preliminaries and Threshold Structure

In this section, we recall the threshold structure of the ℓ_p proximity operator for $0 < p < 1$ and identify the larger stationary branch selected by the global minimizer. Define $\|x\|_p^p = \sum_{i=1}^n |x_i|^p$. We study the proximity operator

$$\text{prox}_{\lambda\|\cdot\|_p^p}(y) = \arg \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda \|x\|_p^p \right\}.$$

By separability, computing $\text{prox}_{\lambda\|\cdot\|_p^p}(y)$ reduces to the scalar problem for $y \in \mathbb{R}$,

$$\text{prox}_{\lambda|\cdot|^p}(y) = \arg \min_{x \in \mathbb{R}} \left\{ \frac{1}{2} (x - y)^2 + \lambda |x|^p \right\}. \quad (1)$$

Define the thresholds

$$\rho_{p,\lambda} := (2\lambda(1-p))^{\frac{1}{2-p}}, \quad \tau_{p,\lambda} := \frac{2-p}{2(1-p)} \rho_{p,\lambda}. \quad (2)$$

According to [12, Propositions 2 and 3],

$$\text{prox}_{\lambda|\cdot|^p}(y) = \begin{cases} 0, & |y| < \tau_{p,\lambda}, \\ \{0, \text{sgn}(y) \rho_{p,\lambda}\}, & |y| = \tau_{p,\lambda}, \\ \text{sgn}(y) x_+(|y|), & |y| > \tau_{p,\lambda}, \end{cases} \quad (3)$$

where $x_+(|y|)$ is the larger positive root of the first-order optimality condition for problem (1), that is,

$$\Phi_{p,\lambda,y}(x) := x - y + \lambda p \text{sgn}(x) |x|^{p-1} = 0. \quad (4)$$

It is known [12, Lemma 3] that for $|y| > \tau_{p,\lambda}$, the function $\Phi_{p,\lambda,y}$ in (4) has exactly two distinct roots $x_-(|y|)$ and $x_+(|y|)$ in $(0, \infty)$, ordered as

$$x_-(|y|) < (\lambda p(1-p))^{\frac{1}{2-p}} < x_+(|y|). \quad (5)$$

Thus, proximal evaluation reduces to the larger root $x_+(|y|)$. Closed forms are known for $p = \frac{1}{2}$ and $p = \frac{2}{3}$ [10, 12]; otherwise Newton or bisection is commonly used [12, 24]. We seek instead a series valid for every $0 < p < 1$.

3 A Series Representation for $x_+(|y|)$

For $p = \frac{1}{2}$ and $p = \frac{2}{3}$, the stationarity equation (4) reduces to a cubic or a structured quartic, respectively, yielding known closed-form solutions [10, 12]. For general p , however, the algebraic degree grows rapidly when p is rational, while the equation is generally nonalgebraic when p is irrational. We therefore reformulate (4) as an inverse-function problem.

Writing the stationarity equation as $u = f(v)$, with f analytic near a regular point, the Lagrange–Bürmann theorem provides a convergent series for the inverse branch $v = g(u)$ [15, 18]. Applied here, it yields a unified series for $x_+(|y|)$ for every $p \in (0, 1)$ and finite hypergeometric representations when p is rational. Some proximity operators also admit closed forms through the Lambert W function—whose series and asymptotics follow from Lagrange inversion—notably for entropic/KL-type penalties and Poisson data terms [3, 14]. We first recall the Lagrange–Bürmann theorem.

Lemma 3.1 (Lagrange–Bürmann [31]). *Let f be analytic in a neighborhood of $v_0 \in \mathbb{C}$ with $f(v_0) = u_0$ and $f'(v_0) \neq 0$. Then f admits a local analytic inverse g near u_0 with $g(u_0) = v_0$, and, for u near u_0 ,*

$$g(u) = v_0 + \sum_{n=1}^{\infty} \frac{(u - u_0)^n}{n!} \frac{d^{n-1}}{dw^{n-1}} \left(\left(\frac{w - v_0}{f(w) - u_0} \right)^n \right) \Big|_{w=v_0}. \quad (6)$$

Thus, if f is locally one-to-one at v_0 , its inverse can be expanded at $u_0 = f(v_0)$ using derivatives of f , without solving $u = f(v)$ in closed form.

To apply the Lagrange–Bürmann inversion theorem to the ℓ_p proximal problem, we recall the first-order optimality condition (4):

$$\Phi_{p,\lambda,y}(x) := x - y + \lambda p \operatorname{sgn}(x) |x|^{p-1} = 0.$$

Figure 1 illustrates $\Phi_{p,\lambda,y}(x)$ for $p = \frac{2}{3}$, $\lambda = 0.8$, and $y = 1.5$; the qualitative shape is similar for other choices of (p, λ, y) . For $y > 0$, any root of $\Phi_{p,\lambda,y}$ lies in $(0, \infty)$ (in fact, from $x - y + \lambda p x^{p-1} = 0$ we obtain $x = y - \lambda p x^{p-1} \in (0, y)$), so we restrict attention to $x > 0$.

To put the equation in an invertible analytic form suitable for Lagrange–Bürmann, divide by $y > 0$ and set

$$\varepsilon := \lambda p y^{p-2}, \quad v := \frac{x}{y}, \quad f(v) := (1 - v) v^{1-p}.$$

Then (4) is equivalent to the scalar equation

$$f(v) = \varepsilon.$$

Thus, solving $\Phi_{p,\lambda,y}(x) = 0$ is equivalent to inverting the analytic map $v \mapsto f(v)$ at the value ε , with the larger root corresponding to the branch $v \in (0, 1)$ near $v = 1$. (Note that f is analytic near $v = 1$ and $f'(1) = -1 \neq 0$, so the Lagrange–Bürmann inversion applies.)

On $(0, \infty)$, $\Phi_{p,\lambda,y}$ has a unique minimum, whereas f has the unique maximizer $v_\star = (1 - p)/(2 - p)$. When two stationary roots exist, the right intersection $v_+ \in (v_\star, 1)$ maps to the larger root $x_+(y) = yv_+$. Figure 1 illustrates these equivalent descriptions.

Lemma 3.2 (Local inverse and radius of convergence). *Let $0 < p < 1$. Set $\mathbb{D} := \mathbb{C} \setminus (-\infty, 0]$ and let $\operatorname{Log} : \mathbb{D} \rightarrow \mathbb{C}$ denote the principal branch with $\operatorname{Log}(1) = 0$. Define*

$$f(v) := (1 - v)v^{1-p}, \quad v^\alpha := \exp(\alpha \operatorname{Log} v) \quad (\alpha \in \mathbb{R}).$$

Then there exist neighborhoods $U \ni 0$ and $V \ni 1$, with $V \subset \mathbb{D}$, such that $f : V \rightarrow U$ is biholomorphic and its inverse $g : U \rightarrow V$ satisfies $g(0) = 1$. Moreover, the Taylor series of g is

$$g(u) = 1 - u - \sum_{n=2}^{\infty} \frac{u^n}{n!} \cdot \prod_{j=0}^{n-2} (n(1 - p) + j). \quad (7)$$

and has radius of convergence $u_\star$, where

$$v_\star := \frac{1 - p}{2 - p}, \quad u_\star := f(v_\star) = \frac{(1 - p)^{1-p}}{(2 - p)^{2-p}}. \quad (8)$$

Here, $v_\star$ is the unique maximizer of f on $(0, 1)$ and $u_\star$ is its maximum value.

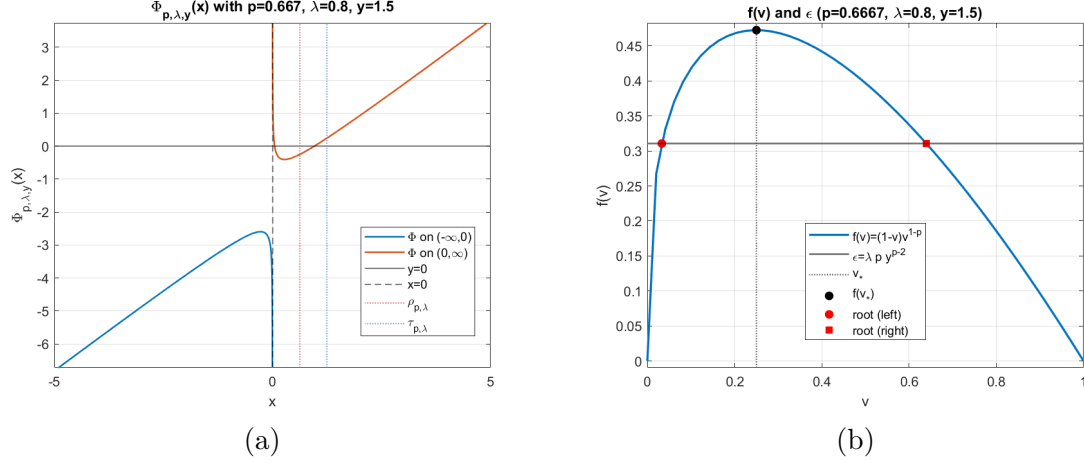


Figure 1: **Stationary equation and reparameterization.** (a) Graph of $\Phi_{p,\lambda,y}(x)$ for $p = \frac{2}{3}$, $\lambda = 0.8$, and $y = 1.5$. On $(0, \infty)$ the curve decreases to a unique minimum at $x_{\min} = (\lambda p(1-p))^{1/(2-p)}$ and then increases, producing two positive roots $x_-(y) < x_+(y)$ when the minimum is negative. (b) The transformed equation $f(v) = (1-v)v^{1-p} = \epsilon$ with $\epsilon = \lambda p y^{p-2}$ on $v \in [0, 1]$. Intersections $v_{\pm}$ map to the zeros of $\Phi_{p,\lambda,y}$ by $x_{\pm} = y v_{\pm}$; the right intersection v_+ corresponds to the larger root $x_+(y)$ used in the proximal operator. The function f attains its maximum at $v_{\star} = \frac{1-p}{2-p}$; the critical level $\epsilon = f(v_{\star})$ marks the transition between two, one (merged), and zero intersections.

Proof. The function f is analytic at $v_0 = 1$, with $u_0 = f(v_0) = 0$ and $f'(v_0) = -1 \neq 0$. By Lemma 3.1, the local inverse g of f at 0 satisfies

$$g(u) = 1 + \sum_{n=1}^{\infty} \frac{u^n}{n!} \frac{d^{n-1}}{dw^{n-1}} \left(\left(\frac{w-1}{f(w)} \right)^n \right) \Big|_{w=1}.$$

For $f(w) = (1-w)w^{1-p}$, $\left(\frac{w-1}{f(w)} \right)^n = (-w^{p-1})^n = (-1)^n w^{n(p-1)}$, hence

$$\frac{d^{n-1}}{dw^{n-1}} \left(\left(\frac{w-1}{f(w)} \right)^n \right) \Big|_{w=1} = - \prod_{j=0}^{n-2} (n(1-p) + j),$$

for $n \geq 2$. This proves (7).

To determine the radius, note first that $f'(v) = -v^{1-p} + (1-v)(1-p)v^{-p}$, so f' vanishes only at $v_{\star}$ and $u_{\star}$ is a critical value of f .

Fix the open disk

$$B := \{v : |v-1| < \rho\}, \quad \rho := 1 - v_{\star}.$$

Then f is analytic on $\bar{B}$ and $f'(v) \neq 0$ on B (the only critical point $v_{\star}$ lies on ∂B). On ∂B , write $v = 1 - \rho e^{i\theta}$. A direct estimate gives

$$\min_{v \in \partial B} |f(v)| = \min_{\theta \in [0, 2\pi]} \rho |1 - \rho e^{i\theta}|^{1-p} = \rho(1-\rho)^{1-p} = (1-v_{\star})v_{\star}^{1-p} = u_{\star}.$$

Therefore, for any $|\tilde{u}| < u_{\star}$, Rouché's theorem shows that f and $f - \tilde{u}$ have the same number of zeros in B . Since f has the unique zero $v = 1$ in B , the equation $f(v) = \tilde{u}$ also has a unique solution $\tilde{v} \in B$. Because $f'(\tilde{v}) \neq 0$, the inverse g is analytic at $\tilde{u}$, with $g(\tilde{u}) = \tilde{v}$. Hence g is analytic on $\{u : |u| < u_{\star}\}$.

Finally, u_* is a singular value for the inverse: the two real preimages $v_{\pm}$ coalesce at v_* , where $f'(v_*) = 0$ and $f''(v_*) = -(2-p)v_*^{-p} \neq 0$. Thus the inverse has a square-root branch point at u_* , so the radius of convergence is exactly u_* . $\square$

Next we present a series representation and prove that it converges to the larger stationary root throughout the active proximal domain.

Theorem 3.3 (Series for the larger root). *Let $0 < p < 1$, $\lambda > 0$, and $y \geq \tau_{p,\lambda}$. Define*

$$\varepsilon(y) := \lambda p y^{p-2},$$

$$S(y) := y \left(1 - \varepsilon(y) - \sum_{n=2}^{\infty} \frac{\varepsilon(y)^n}{n!} \prod_{j=0}^{n-2} (n(1-p) + j) \right).$$

The series defining $S(y)$ is absolutely convergent, uniformly for $y \in [\tau_{p,\lambda}, \infty)$. Moreover, $S(y) = x_+(y)$, the larger positive solution of the stationary equation in (4).

Proof. Let f and its inverse branch g be as in Lemma 3.2. Since $\varepsilon(y)$ decreases with y , for $y \geq \tau_{p,\lambda}$,

$$\varepsilon(y) \leq \varepsilon(\tau_{p,\lambda}) = q_p u_* < u_*, \quad q_p := p 2^{1-p} < 1.$$

Indeed, $p \mapsto \log(p 2^{1-p})$ is strictly increasing on $(0, 1)$ and vanishes at $p = 1$. The power series for g therefore converges absolutely and uniformly for all values of $\varepsilon(y)$ in the compact interval $[0, q_p u_*]$. The same remains true after rescaling by y . Indeed, for $n \geq 1$,

$$y \varepsilon(y)^n = (\lambda p)^n y^{1-n(2-p)}$$

decreases on $[\tau_{p,\lambda}, \infty)$. Thus the tail of the series for $S(y)$ is bounded by $\tau_{p,\lambda}$ times the corresponding tail at $\varepsilon(\tau_{p,\lambda})$, which tends to zero uniformly. Set $z(y) := g(\varepsilon(y))$ and $x(y) := yz(y)$. Since $f(z(y)) = \varepsilon(y)$, the definition of f shows that $\Phi_{p,\lambda,y}(x(y)) = 0$, and substituting (7) gives the displayed formula for $S(y)$.

It remains to identify the branch. For each $0 < \varepsilon < u_*$, the equation $f(v) = \varepsilon$ has two real solutions, one in $(0, v_*)$ and one in $(v_*, 1)$. Because $g(0) = 1$, continuity gives $g(\varepsilon) \in (v_*, 1)$; hence it selects the larger solution. Therefore $x(y) = x_+(y)$ throughout the active domain. $\square$

To conclude this section, we record a compact Gamma-function form of $S(y)$. Using $\Gamma(z+1) = z\Gamma(z)$, we may write

$$S(y) = y \left(1 - \sum_{n=1}^{\infty} \frac{\varepsilon(y)^n}{n!} \frac{\Gamma(n(2-p) - 1)}{\Gamma(n(1-p))} \right), \tag{9}$$

where $\varepsilon(y) := \lambda p y^{p-2}$. Denote the parenthesized quantity by $z(\varepsilon)$. Then $S(y) = yz(\varepsilon(y))$.

4 Rational Exponents and Hypergeometric Representations

Let $p = r/s \in (0, 1)$, where r and s are coprime positive integers with $r < s$. For $x > 0$, set $t = x^{1/s}$. Multiplying the stationarity equation

$$x - |y| + \lambda p x^{p-1} = 0$$

by $x^{1-p} = t^{s-r}$ gives

$$t^{2s-r} - |y|t^{s-r} + \lambda p = 0,$$

which is a trinomial algebraic equation. Thus, the existence of special-function representations in the rational case belongs to the general theory of trinomial roots [17, 20, 26]. The purpose of this section is to specialize the unified inverse series derived above to the real proximal branch and give its residue-class decomposition explicitly. The resulting formula fixes both the relevant branch and its domain of validity. We then recover the familiar formulas for $p = \frac{1}{2}$ and $p = \frac{2}{3}$ and, to the best of our knowledge, obtain a compact $p = \frac{1}{3}$ representation not previously recorded in the proximal-operator literature.

Let us first introduce the rising Pochhammer symbol and the hypergeometric function that will simplify our discussion.

We use

$$(a)_n := \overbrace{a(a+1)(a+2) \cdots (a+n-1)}^{n \text{ factors}}$$

for the rising Pochhammer symbol, with $(a)_0 = 1$; equivalently, $(a)_n = \Gamma(a+n)/\Gamma(a)$.

The generalized hypergeometric function associated with two sequences $(a_i)_{i=1}^\alpha$ and $(b_i)_{i=1}^\beta$ is defined by the power series

$${}_aF_\beta(a_1, \dots, a_\alpha; b_1, \dots, b_\beta; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_\alpha)_n}{(b_1)_n \cdots (b_\beta)_n} \frac{z^n}{n!}.$$

The radius of convergence of ${}_aF_\beta$ is as follows [25]:

- If $\alpha \leq \beta$, then ${}_aF_\beta$ is entire (*infinite* radius of convergence).
- If $\alpha = \beta + 1$, the radius is *one* ($|z| < 1$).
- If $\alpha \geq \beta + 2$, the radius is *zero* (diverges for all $z \neq 0$, unless the series terminates).

We repeatedly use Gauss's multiplication formula [1],

$$\prod_{j=0}^{M-1} \Gamma\left(z + \frac{j}{M}\right) = (2\pi)^{\frac{M-1}{2}} M^{\frac{1}{2}-Mz} \Gamma(Mz). \quad (10)$$

Proposition 4.1 (Explicit rational-exponent decomposition). *Let $p = r/s \in (0, 1)$, where r and s are coprime, and set*

$$A := 2s - r, \quad B := s - r, \quad \kappa := \frac{A^A}{B^B s^s}.$$

For $j = 1, \dots, s$, define

$$\alpha_j := \frac{Aj}{s} - 1, \quad \beta_j := \frac{Bj}{s}, \quad C_j := \frac{\Gamma(\alpha_j)}{\Gamma(\beta_j)\Gamma(j+1)},$$

and the parameter multisets

$$\begin{aligned} \mathcal{A}_j &:= \left\{ \frac{\alpha_j + k}{A} : k = 0, \dots, A-1 \right\}, \\ \mathcal{B}_j &:= \left(\left\{ \frac{\beta_j + k}{B} : k = 0, \dots, B-1 \right\} \cup \left\{ \frac{j+1+k}{s} : k = 0, \dots, s-1 \right\} \right) \setminus \{1\}, \end{aligned}$$

where one occurrence of 1 is removed. Then, for $|\varepsilon| < u_\star$,

$$z(\varepsilon) = 1 - \sum_{j=1}^s C_j \varepsilon^j {}_A F_{A-1}(\mathcal{A}_j; \mathcal{B}_j; \kappa \varepsilon^s). \quad (11)$$

Moreover, $\kappa u_\star^s = 1$, so each hypergeometric argument has modulus less than one on the disk of convergence of z .

Proof. Write each $n \geq 1$ uniquely as $n = sm + j$, with $m \geq 0$ and $j \in \{1, \dots, s\}$. The coefficient in (9) is

$$\frac{\Gamma(Am + \alpha_j)}{\Gamma(Bm + \beta_j) \Gamma(sm + j + 1)}.$$

Applying Gauss's multiplication formula (10) to its three Gamma factors and using $\Gamma(m + \theta) = \Gamma(\theta)(\theta)_m$ shows that its ratio to C_j is

$$\kappa^m \frac{\prod_{a \in \mathcal{A}_j} (a)_m}{(1)_m \prod_{b \in \mathcal{B}_j} (b)_m}.$$

Indeed, the two denominator parameter multisets contain exactly one occurrence of 1, which supplies $(1)_m = m!$. Summing first over m and then over j gives (11). Finally, (8) gives $u_\star^s = B^B s^s / A^A = \kappa^{-1}$. $\square$

Following this construction, we give three illustrative examples: $p = \frac{1}{2}$, $p = \frac{2}{3}$, and $p = \frac{1}{3}$.

4.1 The Case $p = \frac{1}{2}$

Theorem 4.2 (Hypergeometric form for $p = \frac{1}{2}$). *For $\lambda > 0$ and $y \geq \tau_{1/2, \lambda} = \frac{3}{2} \lambda^{2/3}$, set $\varepsilon = \frac{1}{2} \lambda y^{-3/2}$ and $t = \frac{27}{4} \varepsilon^2$. Then the function $S(y)$ defined in Theorem 3.3 can be written as*

$$S(y) = y \left(\frac{2}{3} + \frac{1}{3} {}_2F_1 \left(-\frac{1}{3}, \frac{1}{3}; \frac{1}{2}; t \right) - \varepsilon {}_2F_1 \left(\frac{1}{6}, \frac{5}{6}; \frac{3}{2}; t \right) \right). \quad (12)$$

Proof. For $p = \frac{1}{2}$, write

$$a_n := \frac{\Gamma(3n/2 - 1)}{n! \Gamma(n/2)}, \quad \frac{S(y)}{y} = 1 - \sum_{n \geq 1} a_n \varepsilon^n.$$

Splitting the sum into even and odd indices and applying (10) gives

$$a_{2m} = -\frac{1}{3} \left(\frac{27}{4} \right)^m \frac{(-\frac{1}{3})_m (\frac{1}{3})_m}{(\frac{1}{2})_m m!}, \quad a_{2m+1} = \left(\frac{27}{4} \right)^m \frac{(\frac{1}{6})_m (\frac{5}{6})_m}{(\frac{3}{2})_m m!}.$$

The two identities hold for $m \geq 1$ and $m \geq 0$, respectively. Substitution and $t = 27\varepsilon^2/4$ yield (12). Finally, $y \geq \tau_{1/2, \lambda}$ implies $\varepsilon \leq \sqrt{2}/(3\sqrt{3})$, hence $0 \leq t \leq \frac{1}{2}$, so both Gauss series converge absolutely. $\square$

The standard identities for ${}_2F_1(-a, a; \frac{1}{2}; \sin^2 \phi)$ and ${}_2F_1(a, 1-a; \frac{3}{2}; \sin^2 \phi)$ [1, Eqs. 15.1.15 and 15.1.17], applied with $\phi = \arcsin \sqrt{t}$, reduce (12) to the following trigonometric form.

Corollary 4.3. *Under the conditions of Theorem 4.2, set $\phi = \arcsin \sqrt{t}$. Then*

$$S(y) = y \left(\frac{2}{3} + \frac{1}{3} \cos\left(\frac{2}{3}\phi\right) - \frac{\sqrt{3}}{3} \sin\left(\frac{2}{3}\phi\right) \right) = \frac{2y}{3} \left(1 + \cos\left(\frac{2\phi}{3} + \frac{\pi}{3}\right) \right). \quad (13)$$

Consequently, for $y \geq \tau_{1/2, \lambda}$, the larger positive root $x_+(y)$ coincides with the expression for $S(y)$ in Corollary 4.3.

4.2 The Cases $p = \frac{2}{3}$ and $p = \frac{1}{3}$

Theorem 4.4 (Hypergeometric form for $p = \frac{2}{3}$). For $\lambda > 0$ and $y \geq \tau_{2/3,\lambda} = 2(\frac{2}{3}\lambda)^{3/4}$, set

$$\varepsilon = \frac{2}{3}\lambda y^{-4/3}, \quad t = \frac{256}{27}\varepsilon^3,$$

and let $S(y)$ be as in Theorem 3.3. Then

$$\begin{aligned} \frac{S(y)}{y} &= \frac{3}{4} + \frac{1}{4} {}_3F_2\left(\frac{1}{2}, -\frac{1}{4}, \frac{1}{4}; \frac{1}{3}, \frac{2}{3}; t\right) \\ &\quad - \varepsilon {}_3F_2\left(\frac{1}{12}, \frac{7}{12}, \frac{5}{6}; \frac{2}{3}, \frac{4}{3}; t\right) - \frac{\varepsilon^2}{3} {}_3F_2\left(\frac{5}{12}, \frac{11}{12}, \frac{7}{6}; \frac{4}{3}, \frac{5}{3}; t\right). \end{aligned} \quad (14)$$

and the series converges absolutely for $0 \leq t \leq 16/27$ whenever $y \geq \tau_{2/3,\lambda}$.

Theorem 4.5 (Hypergeometric form for $p = \frac{1}{3}$). Let $\lambda > 0$ and $y \geq \tau_{1/3,\lambda}$. Set

$$\varepsilon = \frac{1}{3}\lambda y^{-5/3}, \quad t = \frac{3125}{108}\varepsilon^3. \quad (15)$$

Then

$$\begin{aligned} \frac{S(y)}{y} &= \frac{3}{5} + \frac{2}{5} {}_4F_3\left(-\frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}; \frac{1}{3}, \frac{1}{2}, \frac{2}{3}; t\right) \\ &\quad - \varepsilon {}_4F_3\left(\frac{2}{15}, \frac{8}{15}, \frac{11}{15}, \frac{14}{15}; \frac{2}{3}, \frac{5}{6}, \frac{4}{3}; t\right) \\ &\quad - \frac{2}{3}\varepsilon^2 {}_4F_3\left(\frac{7}{15}, \frac{13}{15}, \frac{16}{15}, \frac{19}{15}; \frac{7}{6}, \frac{4}{3}, \frac{5}{3}; t\right). \end{aligned} \quad (16)$$

Moreover, at the threshold $y = \tau_{1/3,\lambda}$ one has $t = \frac{4}{27} < 1$, so for all $y \geq \tau_{1/3,\lambda}$ the argument $t \in (0, \frac{4}{27}]$ and all ${}_4F_3$ series converge absolutely.

Proof. For both theorems, write

$$a_n := \frac{\Gamma(n(2-p)-1)}{n!\Gamma(n(1-p))}, \quad z(\varepsilon) = 1 - \sum_{n \geq 1} a_n \varepsilon^n.$$

For $p = \frac{2}{3}$, split the sum according to $n = 3m, 3m+1$, and $3m+2$. Gauss's multiplication formula gives, for $m \geq 0$ (and for $m \geq 1$ in the first line),

$$\begin{aligned} -a_{3m} &= \frac{1}{4} \frac{(\frac{1}{2})_m (-\frac{1}{4})_m (\frac{1}{4})_m}{(\frac{1}{3})_m (\frac{2}{3})_m m!} \left(\frac{256}{27}\right)^m, \\ a_{3m+1} &= \frac{(\frac{1}{12})_m (\frac{7}{12})_m (\frac{5}{6})_m}{(\frac{2}{3})_m (\frac{4}{3})_m m!} \left(\frac{256}{27}\right)^m, \\ a_{3m+2} &= \frac{1}{3} \frac{(\frac{5}{12})_m (\frac{11}{12})_m (\frac{7}{6})_m}{(\frac{4}{3})_m (\frac{5}{3})_m m!} \left(\frac{256}{27}\right)^m. \end{aligned}$$

Substitution into the three residue-class sums proves Theorem 4.4.

For $p = \frac{1}{3}$, the same calculation yields

$$\begin{aligned} -a_{3m} &= \frac{2}{5} \frac{(-\frac{1}{5})_m (\frac{1}{5})_m (\frac{2}{5})_m (\frac{3}{5})_m}{(\frac{1}{3})_m (\frac{1}{2})_m (\frac{2}{3})_m m!} \left(\frac{3125}{108} \right)^m, \\ a_{3m+1} &= \frac{(\frac{2}{15})_m (\frac{8}{15})_m (\frac{11}{15})_m (\frac{14}{15})_m}{(\frac{2}{3})_m (\frac{5}{6})_m (\frac{4}{3})_m m!} \left(\frac{3125}{108} \right)^m, \\ a_{3m+2} &= \frac{2}{3} \frac{(\frac{7}{15})_m (\frac{13}{15})_m (\frac{16}{15})_m (\frac{19}{15})_m}{(\frac{7}{6})_m (\frac{4}{3})_m (\frac{5}{3})_m m!} \left(\frac{3125}{108} \right)^m. \end{aligned}$$

These identities give (16). The stated ranges follow by substituting the activation thresholds into t ; all displayed hypergeometric series have radius one. The proof is complete. $\square$

MATLAB's built-in function `hypergeom` evaluates the generalized hypergeometric functions ${}_pF_q$ and can be used to compute the expressions above.

Although these special-function representations exist for all rational exponents, formulas by radicals are much more restrictive. A separate supplement records the corresponding Galois-theoretic classification.

5 Mellin–Barnes Representation and Error-Controlled Evaluation

In this section, we recast the larger proximal root through the normalized map $\varepsilon(y) := \lambda p y^{p-2}$ and $z(\varepsilon) := x_+(y)/y$. We derive a Mellin–Barnes (MB) integral for z , show that a contour shift reproduces a finite part of the Lagrange–Bürmann series, and bound the remaining vertical tail. The mathematical truncation error and the numerical quadrature error are recorded separately. This distinction is essential: the asymptotic estimates below prove uniform analytic bounds, whereas a formally certified floating-point implementation would additionally require explicit Gamma-function constants and validated quadrature.

5.1 Mellin–Barnes Representation and Equivalence to the Series

In this subsection, we derive a Mellin–Barnes (MB) integral for $z(\varepsilon)$ and show that closing the contour to the right recovers the Lagrange–Bürmann series.

Lemma 5.1 (Semicircle sine bound). *Let $0 < p < 1$ and $\gamma \in (\frac{1}{2-p}, 1)$. For $R > 0$ and $\theta \in [-\pi/2, \pi/2]$, set*

$$s = \gamma + Re^{i\theta} = \Re s + i \Im s, \quad \Re s = \gamma + R \cos \theta, \quad \Im s = R \sin \theta.$$

If $R = R_k := k + \frac{1}{2} - \gamma$ with $k \in \mathbb{N}$ sufficiently large, then

$$\frac{1}{|\sin(\pi s)|} \leq \begin{cases} 2, & \text{if } |\theta| \leq R^{-1}, \\ \frac{2}{1-e^{-4}} e^{-\pi |\Im s|}, & \text{if } R^{-1} \leq |\theta| \leq \frac{\pi}{2}. \end{cases}$$

Proof. Use the identity $|\sin(\pi s)|^2 = \sin^2(\pi \Re s) + \sinh^2(\pi \Im s)$, fix large k , and write $R = R_k$. We consider separately $|\theta| \leq R^{-1}$ and $R^{-1} \leq |\theta| \leq \pi/2$.

Small angles. Since $\cos \theta \geq 1 - \theta^2/2$, we have

$$\Re s = \gamma + R \cos \theta \geq \gamma + R \left(1 - \frac{\theta^2}{2} \right) = \left(k + \frac{1}{2} \right) - \frac{R\theta^2}{2}.$$

Let $\delta := (k + \frac{1}{2}) - \Re s \in [0, \frac{R\theta^2}{2}]$. If $|\theta| \leq R^{-1}$ then $\delta \leq \frac{R\theta^2}{2} \leq \frac{R}{2} R^{-2} = \frac{1}{2R}$. Hence

$$|\sin(\pi \Re s)| = |\sin(\pi(k + \frac{1}{2} - \delta))| = \cos(\pi\delta) \geq \cos\left(\frac{\pi}{2R}\right) \geq \frac{1}{2}$$

for all large R . Therefore,

$$|\sin(\pi s)| \geq |\sin(\pi \Re s)| \geq \frac{1}{2}, \quad \frac{1}{|\sin(\pi s)|} \leq 2,$$

in the first range.

Remaining angles. Using $\sin|\theta| \geq \frac{2}{\pi}|\theta|$ when $|\theta| \leq \frac{\pi}{2}$,

$$|\Im s| = R|\sin \theta| \geq R \cdot \frac{2}{\pi}|\theta| \geq \frac{2}{\pi}.$$

Consequently,

$$\sinh(\pi|\Im s|) = \frac{e^{\pi|\Im s|}}{2}(1 - e^{-2\pi|\Im s|}) \geq \frac{1 - e^{-4}}{2}e^{\pi|\Im s|}.$$

Hence, $\frac{1}{\sinh \pi|\Im s|} \leq \frac{2}{1 - e^{-4}}e^{-\pi|\Im s|}$. Combining the two cases yields the claim. $\square$

Lemma 5.2 (Uniform bound for $G(s)$). *Fix $0 < p < 1$ and $\gamma \in (\frac{1}{2-p}, 1)$. Define*

$$G(s) := \frac{\Gamma((2-p)s - 1)}{\Gamma(1+s)\Gamma((1-p)s)}, \quad A := (2-p)\log(2-p) - (1-p)\log(1-p).$$

For the right semicircle

$$\Sigma_R := \{ \gamma + R \cos \theta + iR \sin \theta : \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \},$$

there exist constants $C > 0$ and $R_0 > 0$ (depending only on p, γ) such that for all $R \geq R_0$ and all $s \in \Sigma_R$,

$$|G(s)| \leq C R^{-3/2} e^{A \Re s}. \quad (17)$$

Proof. Stirling's formula, uniformly for the three Gamma-function arguments on the right half-plane under consideration, gives

$$G(s) = C_p e^{As} s^{-3/2} (1 + O(|s|^{-1})) \quad (|s| \rightarrow \infty),$$

where $C_p \neq 0$ depends only on p . On Σ_R ,

$$|s| = |\gamma + Re^{i\theta}| \geq R - \gamma \geq R/2$$

once $R \geq 2\gamma$. Taking absolute values and enlarging a constant to cover the bounded remaining range proves (17). $\square$

Armed with the two lemmas above, we are ready to establish the main result of this subsection.

Theorem 5.3 (Mellin–Barnes representation). *Let $0 < p < 1$ and $0 < \varepsilon < u_\star := (1-p)^{1-p}/(2-p)^{2-p}$. Define*

$$z(\varepsilon) := 1 - \sum_{n=1}^{\infty} \frac{\varepsilon^n}{n!} \frac{\Gamma(n(2-p) - 1)}{\Gamma(n(1-p))}.$$

Then for any $\gamma \in (\frac{1}{2-p}, 1)$,

$$z(\varepsilon) = 1 - \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} \Gamma(-s) \frac{\Gamma(s(2-p)-1)}{\Gamma(s(1-p))} (-\varepsilon)^s ds, \quad (18)$$

where

$$(-\varepsilon)^s := \exp(s(\log \varepsilon + i\pi))$$

denotes the upper boundary value on the negative real axis.

Proof. Let

$$f(s) := \Gamma(-s) \frac{\Gamma(s(2-p)-1)}{\Gamma(s(1-p))} (-\varepsilon)^s.$$

The poles of (f) arise from the following two gamma factors:

$$\begin{aligned} \Gamma(-s) : \quad s &= 0, 1, 2, \dots, \\ \Gamma(s(2-p)-1) : \quad s &= \frac{1}{2-p}, 0, -\frac{1}{2-p}, -\frac{2}{2-p}, \dots \end{aligned}$$

Fix $\gamma \in (\frac{1}{2-p}, 1)$ and let $L_{\gamma,R}$ be the vertical segment

$$L_{\gamma,R} := \{\gamma + it : -R \leq t \leq R\}.$$

For $R = R_k$ as in Lemma 5.1, let

$$\Sigma_R = \{ \gamma + R \cos \theta + iR \sin \theta : \theta \in [\frac{\pi}{2}, -\frac{\pi}{2}] \}.$$

Orient $L_{\gamma,R}$ from bottom to top. Then $L_{\gamma,R} \cup \Sigma_R$ is clockwise and encloses the simple poles $s = 1, 2, \dots, \lfloor \gamma + R \rfloor$ of $\Gamma(-s)$, but neither 0 nor $1/(2-p)$.

At $s = n \in \{1, 2, \dots\}$, $\text{Res}_{s=n} \Gamma(-s) = (-1)^{n+1}/n!$, and the other factors are analytic, so

$$\text{Res}_{s=n} f(s) = -\frac{\varepsilon^n}{n!} \frac{\Gamma(n(2-p)-1)}{\Gamma(n(1-p))}.$$

Because the contour is clockwise, the residue theorem introduces a minus sign. Combining it with the negative residues above gives

$$\frac{1}{2\pi i} \int_{L_{\gamma,R} \cup \Sigma_R} f(s) ds = \sum_{n=1}^{\lfloor \gamma+R \rfloor} \frac{\varepsilon^n}{n!} \frac{\Gamma(n(2-p)-1)}{\Gamma(n(1-p))}. \quad (19)$$

Here we write $s = \Re s + i\Im s = (\gamma + R \cos \theta) + iR \sin \theta$. Using the reflection formula $\Gamma(-s)\Gamma(1+s) = -\frac{\pi}{\sin \pi s}$ gives

$$f(s) = -\frac{\pi}{\sin \pi s} \frac{\Gamma(s(2-p)-1)}{\Gamma(1+s)\Gamma(s(1-p))} (-\varepsilon)^s.$$

For simplicity, let $G(s) := \frac{\Gamma(s(2-p)-1)}{\Gamma(1+s)\Gamma(s(1-p))}$. By Lemma 5.2 there exists a real constant $C > 0$ such that

$$|G(s)| \leq CR^{-\frac{3}{2}} e^{A\Re s}.$$

Moreover,

$$|(-\varepsilon)^s| = |e^{s(\log \varepsilon + i\pi)}| = e^{\Re s \log \varepsilon - \pi \Im s}.$$

We now combine these estimates with Lemma 5.1. Set $A := (2-p) \log(2-p) - (1-p) \log(1-p) = -\log u_\star$ and $\tilde{A} := A + \log \varepsilon = \log(\varepsilon/u_\star) < 0$.

Case (a): $|\theta| \leq R^{-1}$. In this case, $|\Im s| \leq 1$, and hence

$$\begin{aligned} \left| \frac{\pi}{\sin \pi s} \cdot G(s) \cdot (-\varepsilon)^s \right| &\leq (2\pi) \cdot \left(CR^{-\frac{3}{2}} e^{A\Re s} \right) \cdot \left(e^{\Re s \log \varepsilon - \pi \Im s} \right) \\ &\leq (2\pi C e^{\pi + \tilde{A}\gamma}) R^{-\frac{3}{2}} \end{aligned}$$

where $\Re s = \gamma + R \cos \theta \geq \gamma$ and $e^{-\pi \Im s} \leq e^\pi$ were used.

Case (b): $R^{-1} \leq \theta \leq \frac{\pi}{2}$.

$$\begin{aligned} \left| \frac{\pi}{\sin \pi s} \cdot G(s) \cdot (-\varepsilon)^s \right| &\leq \left(\frac{2\pi}{1 - e^{-4}} e^{-\pi |\Im s|} \right) \cdot (CR^{-\frac{3}{2}} e^{A\Re s}) \cdot (e^{\Re s \log \varepsilon - \pi \Im s}) \\ &\leq \left(\frac{2\pi C e^{\tilde{A}\gamma - 4}}{1 - e^{-4}} \right) R^{-\frac{3}{2}} \end{aligned}$$

where we apply $\Re s = \gamma + R \cos \theta \geq \gamma$ and $\Im s = R \sin \theta \geq R(\frac{2}{\pi}\theta) \geq \frac{2}{\pi}$.

Case (c): $-\frac{\pi}{2} \leq \theta \leq -R^{-1}$.

$$\begin{aligned} \left| \frac{\pi}{\sin \pi s} \cdot G(s) \cdot (-\varepsilon)^s \right| &\leq \left(\frac{2\pi C}{1 - e^{-4}} \right) \cdot (e^{-\pi |\Im s| - \pi \Im s}) \cdot R^{-\frac{3}{2}} \cdot e^{\tilde{A}\Re s} \\ &\leq \left(\frac{2\pi C e^{\tilde{A}\gamma}}{1 - e^{-4}} \right) R^{-\frac{3}{2}}. \end{aligned}$$

Here we use $\Re s = \gamma + R \cos \theta \geq \gamma$.

Since $\tilde{A} < 0$, there is a constant $C > 0$ such that, for every $|\theta| \leq \frac{\pi}{2}$,

$$|f(s)| \leq \left| \frac{\pi}{\sin \pi s} \right| \cdot |G(s)| \cdot |(-\varepsilon)^s| \leq CR^{-\frac{3}{2}}.$$

Thus we have

$$\left| \int_{\Sigma_R} f(s) ds \right| \leq \int_{\Sigma_R} |f(s)| |ds| \leq CR^{-\frac{3}{2}} (\pi R) = CR^{-\frac{1}{2}} \rightarrow 0, \quad (R \rightarrow \infty)$$

Letting $R \rightarrow \infty$ along the chosen sequence in (19) gives

$$\frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} f(s) ds = \sum_{n=1}^{\infty} \frac{\varepsilon^n}{n!} \frac{\Gamma(n(2-p) - 1)}{\Gamma(n(1-p))}.$$

Subtracting this identity from 1 gives (18). □

5.2 A Residue-Corrected Error-Controlled Evaluator

We combine a finite residue sum with a truncated vertical MB segment of height $2T$. The resulting representation can reduce finite-order truncation error in some regimes, although its numerical benefit depends on p and the contour parameters. We first bound the analytic truncation in exact arithmetic and then add the quadrature error incurred by an implementation.

For $0 < p < 1$ and $0 < \varepsilon < u_\star$, define

$$a_n := \frac{1}{n!} \frac{\Gamma((2-p)n-1)}{\Gamma((1-p)n)}, \quad H(s) := -\frac{\Gamma(-s)\Gamma(s(2-p)-1)}{\Gamma(s(1-p))}.$$

Then for any $\gamma \in (\frac{1}{2-p}, 1)$, by Theorem 5.3

$$z(\varepsilon) = 1 - \sum_{n=1}^{\infty} a_n \varepsilon^n = 1 + \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} H(s) (-\varepsilon)^s ds, \quad (20)$$

where $(-\varepsilon)^s = \exp(s(\log \varepsilon + i\pi))$ is the upper boundary value.

Fix an integer $N \geq 1$ and choose a shifted vertical line $\Re s = \sigma_N$ with $\sigma_N \in (N, N+1)$. Shifting the contour in (20) to $\Re s = \sigma_N$ and summing the residues at $s = 1, \dots, N$ yields the *exact* identity

$$z(\varepsilon) = 1 - \sum_{n=1}^N a_n \varepsilon^n + \frac{1}{2\pi i} \int_{\sigma_N-i\infty}^{\sigma_N+i\infty} H(s) (-\varepsilon)^s ds. \quad (21)$$

Define the truncated map by cutting the vertical segment at height $T > 0$:

$$\tilde{z}_{N,T}^{\sigma_N}(\varepsilon) := 1 - \sum_{n=1}^N a_n \varepsilon^n + \frac{1}{2\pi i} \int_{\sigma_N-iT}^{\sigma_N+iT} H(s) (-\varepsilon)^s ds. \quad (22)$$

Subtracting (22) from (21) gives the clean error split

$$z(\varepsilon) - \tilde{z}_{N,T}^{\sigma_N}(\varepsilon) = \underbrace{\frac{1}{2\pi i} \left(\int_{\sigma_N+iT}^{\sigma_N+i\infty} + \int_{\sigma_N-i\infty}^{\sigma_N-iT} \right) H(s) (-\varepsilon)^s ds}_{E_{\text{MB}}(T, \varepsilon)}. \quad (23)$$

Lemma 5.4 (Series-tail bound). *Set $E_{\text{series}}(N, \varepsilon) := -\sum_{n>N} a_n \varepsilon^n$. There exists $C_2 > 0$, depending only on p , such that for every $0 < r < u_\star$,*

$$\sup_{|\varepsilon| \leq r} |E_{\text{series}}(N, \varepsilon)| \leq C_2 \frac{(r/u_\star)^{N+1}}{1-r/u_\star} \xrightarrow{N \rightarrow \infty} 0. \quad (24)$$

Proof. Stirling's formula, applied to the positive real Gamma-function arguments, gives $a_n = O(n^{-3/2} u_\star^{-n})$. Hence $|a_n| \leq C_2 u_\star^{-n}$ for some $C_2 > 0$ and every $n \geq 1$, after enlarging C_2 to cover finitely many small indices. Thus, for $|\varepsilon| \leq r$, $\sum_{n>N} |a_n| |\varepsilon|^n \leq C_2 \sum_{n>N} (r/u_\star)^n = C_2 \frac{(r/u_\star)^{N+1}}{1-r/u_\star}$. $\square$

Lemma 5.5 (Vertical-tail (MB) bound). *Fix $\sigma_N \in (N, N+1)$ with $N \geq 1$. For any $0 < r < u_\star$ there exists a constant $C' = C'(p, \sigma_N, r)$ such that for all $T > 0$,*

$$\sup_{0 < \varepsilon \leq r} |E_{\text{MB}}(T, \varepsilon)| \leq 2 C' T^{-1/2} \xrightarrow{T \rightarrow \infty} 0.$$

Proof. The standard vertical-line Gamma estimate [28, Eq. 5.11.9]

$$|\Gamma(z)| \leq C \sqrt{2\pi} |\Im z|^{\Re z - 1/2} e^{-\pi |\Im z|/2}, \quad |\Im z| \rightarrow \infty,$$

gives, after enlarging the constant to cover bounded nonzero t , some $C = C(p, \sigma_N) > 0$ such that

$$|H(\sigma_N + it)| \leq C |t|^{-3/2} e^{A\sigma_N} e^{-\pi |t|}, \quad A := -\log u_\star.$$

Using the prescribed upper boundary value, we have

$$|(-\varepsilon)^{\sigma_N + it}| = \varepsilon^{\sigma_N} e^{-\pi t} \leq \varepsilon^{\sigma_N} e^{\pi |t|}.$$

Hence, for $0 < \varepsilon \leq r$,

$$\begin{aligned} |E_{\text{MB}}(T, \varepsilon)| &= \frac{1}{2\pi} \left| \int_{|t| > T} H(\sigma_N + it) (-\varepsilon)^{\sigma_N + it} dt \right| \\ &\leq \frac{C e^{A\sigma_N} r^{\sigma_N}}{2\pi} \int_{|t| > T} |t|^{-3/2} dt \\ &= \frac{2C e^{A\sigma_N} r^{\sigma_N}}{\pi} T^{-1/2} = 2C' T^{-1/2}, \end{aligned}$$

with $C' := \frac{C e^{A\sigma_N} r^{\sigma_N}}{\pi}$, which depends only on (p, σ_N, r) . $\square$

The main result of this subsection is as follows.

Theorem 5.6 (Two-regime analytic truncation). *Fix $0 < p < 1$, $\lambda > 0$, a target accuracy $\eta > 0$, a cap $R \in (0, u_\star)$, and any $\rho \in (0, R)$. Then there exist an integer $N \geq 1$, a shift $\sigma_N \in (N, N+1)$, and a height $T > 0$ such that the piecewise map*

$$\hat{z}(\varepsilon) := \begin{cases} 1 - \sum_{n=1}^N a_n \varepsilon^n, & 0 \leq \varepsilon \leq \rho, \\ 1 - \sum_{n=1}^N a_n \varepsilon^n + \frac{1}{2\pi i} \int_{\sigma_N - iT}^{\sigma_N + iT} H(s) (-\varepsilon)^s ds, & \rho < \varepsilon \leq R, \end{cases} \quad (25)$$

satisfies $\sup_{0 \leq \varepsilon \leq R} |z(\varepsilon) - \hat{z}(\varepsilon)| \leq \eta$. Consequently, for $\varepsilon(y) = \lambda p y^{p-2}$ and $y \geq \tau_{p,\lambda}$ with $\varepsilon(y) \leq R$, the approximation $\hat{x}_+(y) := y \hat{z}(\varepsilon(y))$ obeys $|\hat{x}_+(y) - x_+(y)| \leq y \eta$. In particular, choosing $q_p u_\star \leq R < u_\star$ covers the full active proximal domain.

Proof. (i) *Small- ε branch.* For $0 \leq \varepsilon \leq \rho$, $\hat{z}(\varepsilon) = 1 - \sum_{n=1}^N a_n \varepsilon^n$. Let N be large enough that

$$|z(\varepsilon) - \hat{z}(\varepsilon)| \leq \sum_{n > N} |a_n| \cdot |\varepsilon|^n \leq C_2 \frac{(\rho/u_\star)^{N+1}}{1 - \rho/u_\star} \leq \eta \quad (26)$$

which is possible by Lemma 5.4.

(ii) *MB-augmented branch.* For any $\rho < \varepsilon \leq R$,

$$\hat{z}(\varepsilon) = 1 - \sum_{n=1}^N a_n \varepsilon^n + \frac{1}{2\pi i} \int_{\sigma_N - iT}^{\sigma_N + iT} H(s) (-\varepsilon)^s ds.$$

Use the exact split (23) on $\Re s = \sigma_N \in (N, N+1)$. Choose $T > 0$ so that $2C' T^{-1/2} \leq \eta$ via Lemma 5.5 (with $\sigma = \sigma_N$ and $r = R$). Then for any $\rho < \varepsilon \leq R$, $|z(\varepsilon) - \hat{z}(\varepsilon)| \leq 2C' T^{-1/2} \leq \eta$.

(iii) *Transfer to $x_+(y)$.* Set $\varepsilon(y) = \lambda p y^{p-2}$ and $\hat{x}_+(y) := y \hat{z}(\varepsilon(y))$. Then for any y with $|\varepsilon(y)| \leq R$, $|\hat{x}_+(y) - x_+(y)| = y |\hat{z}(\varepsilon(y)) - z(\varepsilon(y))| \leq y \eta$. $\square$

For numerical use, let $I_{N,T}(\varepsilon)$ denote the finite integral in the second branch of (25). If a validated quadrature returns $\hat{I}_{N,T}(\varepsilon)$ and a bound $E_{\text{quad}}(\varepsilon)$ satisfying

$$|I_{N,T}(\varepsilon) - \hat{I}_{N,T}(\varepsilon)| \leq E_{\text{quad}}(\varepsilon),$$

then replacing $I_{N,T}$ by $\hat{I}_{N,T}$ changes the second-branch bound to

$$|z(\varepsilon) - \hat{z}_{\text{num}}(\varepsilon)| \leq 2C'T^{-1/2} + E_{\text{quad}}(\varepsilon). \quad (27)$$

The constants in Lemmas 5.4 and 5.5 arise from asymptotic Gamma bounds and are not directly available to an implementation. They prove convergence, but a machine-verifiable use of (27) would require explicit numerical majorants and validated quadrature.

An implementable a posteriori control follows instead from the conditioning of the active branch. Define the normalized stationary residual

$$\mathcal{R}_\varepsilon(w) := w - 1 + \varepsilon w^{p-1}.$$

If $w \in [\rho_{p,\lambda}/y, 1]$, then Proposition 5.8 below and the mean-value theorem give

$$|w - z(\varepsilon)| \leq \frac{|\mathcal{R}_\varepsilon(w)|}{1 - p/2}, \quad |yw - x_+(y)| \leq \frac{y|\mathcal{R}_\varepsilon(w)|}{1 - p/2}. \quad (28)$$

Thus the right-hand side supplies a computable exact-arithmetic stopping test for either the series or the MB candidate. A formally certified floating-point result additionally requires directed rounding (or interval arithmetic) for the interval-membership and residual evaluations.

For the truncated series, the coefficients are precomputed and Horner's rule evaluates $1 - \sum_{n=1}^N a_n \varepsilon^n$ in $O(N)$ operations and $O(1)$ auxiliary storage. Algorithm 5.7 combines this evaluation with the residual test.

Algorithm 5.7 (Residual-controlled positive-branch evaluation). Given $p \in (0, 1)$, $\lambda > 0$, $y \geq \tau_{p,\lambda}$, a normalized tolerance $\eta > 0$, a switch $0 < \rho < R < u_\star$, and $\varepsilon(y) \leq R$, proceed as follows.

1. Set $\varepsilon = \lambda p y^{p-2}$, choose an initial order N , and precompute or update the coefficients a_n . For the MB branch also choose $\sigma_N \in (N, N+1)$, $T > 0$, and a quadrature rule.
2. If $\varepsilon \leq \rho$, compute the series candidate $\hat{z} = 1 - \sum_{n=1}^N a_n \varepsilon^n$. Otherwise compute the residue-corrected candidate from (22).
3. If $|\text{Im } \hat{z}|$ exceeds the prescribed arithmetic tolerance, reject the candidate. Otherwise set $w = \text{Re } \hat{z}$, check that $w \in [\rho_{p,\lambda}/y, 1]$, and define

$$\mathcal{E}(y) := \frac{y|\mathcal{R}_\varepsilon(w)|}{1 - p/2}.$$

If $\mathcal{E}(y) \leq y\eta$, return $\hat{x}_+(y) = yw$. Otherwise increase N , or on the MB branch increase T and refine the quadrature, and repeat.

4. If the admissibility test repeatedly fails, use monotone bisection on $[\rho_{p,\lambda}/y, y]$, which is always available on the active branch.

In exact arithmetic, (28) makes $\mathcal{E}(y)$ a rigorous a posteriori bound. Interval evaluation turns the same test into a floating-point certificate.

5.3 Conditioning on the Active Proximal Branch

The activation threshold of the proximity operator must be distinguished from the saddle-node value of the stationary equation. This distinction also clarifies the numerical conditioning of the root that is relevant to the proximal map.

Proposition 5.8 (Uniform conditioning on the active branch). *Let*

$$\rho_{p,\lambda} = (2\lambda(1-p))^{1/(2-p)}, \quad \tau_{p,\lambda} = \frac{2-p}{2(1-p)}\rho_{p,\lambda}.$$

For every $y \geq \tau_{p,\lambda}$, the larger stationary root satisfies $x_+(y) \geq \rho_{p,\lambda}$, and

$$1 - \frac{p}{2} \leq \partial_x \Phi_{p,\lambda,y}(x_+(y)) < 1, \quad 1 < \frac{dx_+(y)}{dy} \leq \frac{1}{1-p/2}.$$

Moreover,

$$0 < \varepsilon(y) \leq q_p u_\star, \quad q_p := p2^{1-p} < 1.$$

Thus, for fixed p , both the inverse sensitivity and the series parameter are uniformly bounded throughout the active proximal domain.

Proof. The saddle-node of the stationary equation occurs at

$$x_{\text{sn}} = (\lambda p(1-p))^{1/(2-p)}, \quad y_{\text{sn}} = \frac{2-p}{1-p}x_{\text{sn}}.$$

It lies strictly below the activation threshold because

$$\frac{\tau_{p,\lambda}}{y_{\text{sn}}} = \frac{1}{2} \left(\frac{2}{p} \right)^{1/(2-p)} > 1.$$

The last inequality is equivalent to $p2^{1-p} < 1$; it also follows because $p \mapsto \log(p2^{1-p})$ is strictly increasing on $(0, 1)$ and has limit zero at $p = 1$.

At activation, the positive global minimizer is $x_+(\tau_{p,\lambda}) = \rho_{p,\lambda}$. The larger root increases with y , so $x_+(y) \geq \rho_{p,\lambda}$. Since $p-2 < 0$,

$$\partial_x \Phi_{p,\lambda,y}(x_+(y)) = 1 - \lambda p(1-p)x_+(y)^{p-2} \geq 1 - \lambda p(1-p)\rho_{p,\lambda}^{p-2} = 1 - \frac{p}{2}.$$

Implicit differentiation gives the asserted bounds on dx_+/dy . Finally, $\varepsilon(y)$ decreases with y , and the calculation in Theorem 3.3 gives $\varepsilon(\tau_{p,\lambda}) = q_p u_\star$. $\square$

In particular, $\Phi_{p,\lambda,y}$ is strictly increasing on $[\rho_{p,\lambda}, y]$, with $\Phi_{p,\lambda,y}(\rho_{p,\lambda}) \leq 0 < \Phi_{p,\lambda,y}(y)$; equality on the left occurs only at activation. For $y > \tau_{p,\lambda}$, the root can therefore be computed robustly by bisection. The series may nevertheless converge more slowly when q_p is close to one, especially as $p \uparrow 1$. The residue-corrected representation can reduce finite-order error, but its contour cost grows in the hardest cases; bisection remains a robust fallback. The difficulty is not a singular active root.

5.4 Certified Use in Proximal Gradient

Let

$$F(x) = h(x) + g(x), \quad g(x) = \lambda \|x\|_p^p,$$

where $h : \mathbb{R}^n \rightarrow \mathbb{R}$ has an L -Lipschitz gradient for some $L > 0$, and let ∂ denote the limiting subdifferential. For $\alpha \in (0, 1/L)$, set

$$Q_\alpha(y; x) := h(x) + \langle \nabla h(x), y - x \rangle + \frac{1}{2\alpha} \|y - x\|^2 + g(y).$$

The following standard implication is recorded to specify the two certificates required of the scalar evaluator.

Theorem 5.9 (Certified inexact proximal gradient). *Assume that F is bounded below and that, for sequences $\delta_k, \eta_k \geq 0$, the computed iterates satisfy*

$$Q_\alpha(x_{k+1}; x_k) \leq F(x_k) + \delta_k, \tag{29}$$

$$r_{k+1} \in \nabla h(x_k) + \frac{1}{\alpha}(x_{k+1} - x_k) + \partial g(x_{k+1}), \quad \|r_{k+1}\| \leq \eta_k. \tag{30}$$

If $\sum_{k=0}^\infty \delta_k < \infty$ and $\eta_k \rightarrow 0$, then, with $c_\alpha := 1/(2\alpha) - L/2 > 0$,

$$\begin{aligned} F(x_{k+1}) &\leq F(x_k) - c_\alpha \|x_{k+1} - x_k\|^2 + \delta_k, \\ \sum_{k=0}^\infty \|x_{k+1} - x_k\|^2 &< \infty, \quad F(x_k) \text{ converges}, \\ \text{dist}(0, \partial F(x_{k+1})) &\leq \eta_k + \left(L + \frac{1}{\alpha}\right) \|x_{k+1} - x_k\| \rightarrow 0. \end{aligned}$$

Consequently, every cluster point of $\{x_k\}$ is a critical point of F .

Proof. The descent lemma gives $F(y) \leq Q_\alpha(y; x) - c_\alpha \|y - x\|^2$, so (29) yields the first inequality. For $A_k := F(x_k) + \sum_{j=k}^\infty \delta_j$, it follows that $A_{k+1} \leq A_k - c_\alpha \|x_{k+1} - x_k\|^2$; hence A_k and $F(x_k)$ converge and the squared steps are summable. Choosing $u_{k+1} \in \partial g(x_{k+1})$ in (30) gives $w_{k+1} := \nabla h(x_{k+1}) + u_{k+1} = r_{k+1} + \nabla h(x_{k+1}) - \nabla h(x_k) - \alpha^{-1}(x_{k+1} - x_k) \in \partial F(x_{k+1})$, and hence the stated distance bound. If $x_{k_j} \rightarrow \bar{x}$, then $x_{k_j+1} \rightarrow \bar{x}$ and $w_{k_j+1} \rightarrow 0$; continuity of F and closedness of ∂F give $0 \in \partial F(\bar{x})$. $\square$

The paper-specific point is that both certificates can be checked from the scalar proximal calculations. Define

$$v_k := x_k - \alpha \nabla h(x_k), \quad \Psi_k(z) := \frac{1}{2\alpha} \|z - v_k\|^2 + g(z).$$

Completing the square gives

$$Q_\alpha(y; x_k) - F(x_k) = \Psi_k(y) - \Psi_k(x_k),$$

so (29) is the directly computable, separable test $\Psi_k(x_{k+1}) \leq \Psi_k(x_k) + \delta_k$. For a nonzero coordinate

$$x_{k+1,i} = \text{sgn}(v_{k,i}) \hat{s}_i$$

the corresponding residual is

$$r_{k+1,i} = \frac{\text{sgn}(v_{k,i})}{\alpha} \Phi_{p,\alpha\lambda,|v_{k,i}|}(\hat{s}_i).$$

At a zero coordinate, $\partial(\lambda|\cdot|^p)(0) = \mathbb{R}$, so the coordinate residual can always be chosen as zero. The residual test alone therefore cannot validate a zero output; this is the role of the model-value test.

Let $\mathcal{I}_k := \{i : x_{k+1,i} \neq 0\}$, and suppose that each $i \in \mathcal{I}_k$ is evaluated on the active branch, with $|v_{k,i}| \geq \tau_{p,\alpha\lambda}$ and $\widehat{s}_i \in [\rho_{p,\alpha\lambda}, |v_{k,i}|]$. Setting $s_i = x_+(|v_{k,i}|)$, Proposition 5.8 and the mean-value theorem give

$$\frac{1-p/2}{\alpha} |\widehat{s}_i - s_i| \leq |r_{k+1,i}| \leq \frac{1}{\alpha} |\widehat{s}_i - s_i|.$$

Thus, an evaluator guarantee $|\widehat{s}_i - s_i| \leq \mathcal{E}_i$ implies $|r_{k+1,i}| \leq \mathcal{E}_i/\alpha$. Since the zero-coordinate residuals may be chosen as above,

$$\|r_{k+1}\| \leq \frac{1}{\alpha} \left(\sum_{i \in \mathcal{I}_k} \mathcal{E}_i^2 \right)^{1/2}.$$

Given prescribed sequences with $\sum_k \delta_k < \infty$ and $\eta_k \rightarrow 0$, the evaluator can therefore be refined until the separable model-value and residual tests meet these tolerances; Theorem 5.9 then applies. This formulation avoids transferring limiting subgradients from an exact proximal point to a nearby approximation, which is not justified by a distance bound alone, and is consistent with standard descent-and-relative-error analyses of nonconvex proximal algorithms [2, 5, 22].

5.5 Numerical Validation

We report two complementary tests: a scalar parameter sweep and an inexact proximal-gradient experiment. Reference scalar roots are computed independently by monotone bisection at 70 and 100 decimal digits. The reported errors are empirical floating-point errors, not validated interval certificates.

For the scalar study, write $z = x_+(y)/y$ and $e_z = |\widehat{z} - z_{\text{ref}}|$. We consider the 135 combinations

$$\begin{aligned} p &\in \{0.1, \tfrac{1}{3}, \tfrac{1}{2}, \tfrac{2}{3}, 0.8, 0.9, 0.95, 0.99, 0.999\}, \\ \lambda &\in \{10^{-6}, 0.8, 10^6\}, \\ y/\tau_{p,\lambda} - 1 &\in \{10^{-10}, 10^{-6}, 10^{-2}, 0.5, 9\}. \end{aligned}$$

Besides fixed truncation orders, an adaptive series doubles N , starting from 16 and stopping at 16384, until its conditioning-based residual indicator is at most 10^{-11} . Table 1(a) shows that the adaptive series meets this target in 129 cases. The six capped cases all have $p = 0.999$ and relative activation gap 10^{-10} or 10^{-6} , confirming that slow series convergence as $p \uparrow 1$, rather than root ill-conditioning, is the difficult regime. Bisection and safeguarded Newton meet the reporting target in all 135 cases.

For a representative residue-corrected calculation with $p = \frac{2}{3}$, $\lambda = 0.8$, and $y = 1.26$, the correction reduces $|\widehat{x} - x_{\text{ref}}|$ from 1.64×10^{-5} to 9.80×10^{-7} at $N = 30$, and from 1.96×10^{-7} to 2.25×10^{-8} at $N = 50$. This illustrates finite-order improvement, not uniform computational superiority. The supplementary $p = 0.99$ tests require increasing contour effort; bisection or safeguarded Newton is preferable in the hardest cases. Detailed contour-height and quadrature tests are provided in the supplementary material.

For the algorithmic test, we minimize $F(x) = \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_p^p$ on two Gaussian 40×70 instances with six-sparse ground truths and noise level 0.01. For $p \in \{0.5, 0.8\}$, we use $x_0 = 0$,

$\lambda = 0.04$, $\alpha = 0.9/\|A\|_2^2$, and the certificates of Theorem 5.9 with $\delta_k = 10^{-12}/(k+1)^2$ and $\eta_k = 10^{-4}/(k+1)^{1.25}$. The stationarity measure is

$$S(x) := \left\| \left((A^\top(Ax - b))_i + \lambda p \operatorname{sgn}(x_i) |x_i|^{p-1} \right)_{i:x_i \neq 0} \right\|_2.$$

All eight series and Newton runs terminate within 177–198 iterations. Across their 1506 accepted steps, the model-value certificate, residual certificate, and predicted descent inequality all hold; the largest observed ratio $\|r_{k+1}\|/\eta_k$ is 2.31×10^{-2} . No Newton fallback is used by the series evaluator. Thus, the computable tests in Section 5.4 can be enforced in a nonconvex proximal-gradient calculation. Full traces and reproducibility details are given in the supplementary material. MATLAB routines for the series and residue-corrected formulas are available in the project repository.

Table 1: Numerical validation. In panel (a), “accurate” means $e_z \leq 10^{-11}$. Panel (b) reports the series-based proximal-gradient runs; safeguarded Newton gives the same K , $F(x_K)$, and $S(x_K)$ to the displayed digits.

(a) Scalar accuracy			(b) Proximal-gradient runs				
Evaluator	Accurate/135	Largest e_z	Seed	p	K	$F(x_K)$	$S(x_K)$
Series, $N = 10$	30	2.13×10^{-2}	1	0.5	177	0.239049	1.62×10^{-7}
Series, $N = 50$	72	4.63×10^{-3}	1	0.8	190	0.248428	1.76×10^{-7}
Series, $N = 110$	90	2.05×10^{-3}	2	0.5	198	0.263414	1.06×10^{-7}
Adaptive series	129	1.00×10^{-7}	2	0.8	188	0.200700	1.09×10^{-7}
Bisection	135	8.82×10^{-12}					
Safeguarded Newton	135	6.79×10^{-12}					

6 Conclusions

We derived an exact Lagrange–Bürmann series for the larger stationary root of the scalar nonconvex ℓ_p proximal problem and proved uniform convergence on the full active proximal domain. For rational p , an explicit residue-class formula reduces the series to a finite sum of generalized hypergeometric functions, including compact formulas for $p \in \{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}\}$. A Mellin–Barnes representation gives an exact residue correction and separates analytic truncation error from quadrature error; an active-branch residual supplies a computable a posteriori stopping test.

The activation threshold is strictly above the saddle–node threshold. As a result, the active proximal root is uniformly well-conditioned; the main near-activation issue is the series convergence factor, particularly when p approaches one. Finally, a model-descent test and a computable stationarity residual yield a rigorous inexact proximal-gradient result: objective values converge, successive steps vanish, and every cluster point is critical under the stated error conditions. The scalar parameter sweep confirms the predicted slowdown as p approaches one, while the sparse-recovery tests show that the two acceptance certificates can be enforced in practice.

Declarations

Funding. The work of Lixin Shen was supported in part by the National Science Foundation under Grant No. DMS-2208385.

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

Data availability. No external datasets were used. The numerical experiments use synthetic data generated from the random seeds specified in the supplementary material.

Code availability. The MATLAB implementation is available at <https://github.com/lshen03/Lp-Proximity-Operator>.

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